

SAFDAR AYUB

AI/ML Engineer | Python Developer | LLM & Automation Specialist

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on production experience building LLM-powered systems, autonomous AI agents, intelligent data pipelines, and automation workflows. Strong Python background across FastAPI, async programming, web scraping (Playwright), PostgreSQL, and cloud-native deployment. Experienced designing and deploying AI solutions independently — from research to production-ready tools. 23+ years of disciplined service in the Pakistan Air Force translating to exceptional research rigor, independent judgment, and delivery under pressure.

TECHNICAL SKILLS

AI & Machine Learning:	OpenAI Agents SDK, Claude Code (Opus), FastMCP, MCP SDK, RAG Systems, Prompt Engineering, Vector Embeddings (Qdrant Cloud), Groq API, Gemini TTS, LLM Orchestration
Python & Frameworks:	Python (Advanced), FastAPI, SQLAlchemy, asyncio, APScheduler, Pydantic, Playwright (web scraping & automation), REST APIs
Databases:	PostgreSQL, SQLite, Neon DB, Turso/LibSQL, Supabase, Qdrant Cloud (vector), Upstash Redis
DevOps & MLOps:	Docker, Kubernetes, Helm Charts, OCI/OKE, GitHub Actions, PM2, NGINX Ingress, Dapr, Apache Kafka, Vercel
Frontend (supporting):	Next.js, React.js, TypeScript, Tailwind CSS, ShadCN UI

KEY PROJECTS

Personal AI Employee — Autonomous AI Agent System ([Hackathon Platinum Tier](#)) | [GitHub](#)

- Built production-grade autonomous AI agent in Python monitoring Gmail, filesystem, and WhatsApp with priority-queue orchestration and human-in-the-loop approval gates
- Implemented 4 custom FastMCP servers (email, social media, Odoo ERP, documents) with circuit breaker patterns and correlation ID tracking
- Designed risk-based routing (routine / sensitive / critical) with dry-run defaults and PM2 process management for reliability
- Used APScheduler for automated task scheduling and Playwright for browser automation and data extraction

Tech: Python, FastMCP, Claude Code, Gmail OAuth2, Playwright, Odoo RPC, APScheduler, PM2

Physical AI & Humanoid Robotics Textbook — RAG Chatbot System | [GitHub](#) | [Live](#)

- Built production RAG pipeline using OpenAI Agents SDK and Qdrant Cloud for semantic similarity search over educational content
- Implemented vector embeddings, intelligent retrieval, and async FastAPI backend with user authentication
- Conducted research across ROS 2, Gazebo, NVIDIA Isaac, and VLA models to produce structured 6-chapter AI curriculum

Tech: FastAPI, OpenAI Agents SDK, MCP SDK, Neon DB, Qdrant Cloud, Better Auth, Vercel

AI-Powered Todo Application — 8-Phase Cloud-Native Evolution | [GitHub](#) | [Live](#)

- Integrated Groq API for natural language intent classification and AI task management with full conversation history
- Evolved system across 8 phases: AI chatbot → Kubernetes → Kafka event streaming → Dapr microservices → OCI Cloud
- Deployed production system on OCI OKE with Helm charts, NGINX Ingress, and Dapr sidecar injection

Tech: FastAPI, SQLAlchemy, Groq API, Docker, Kubernetes, Helm 3, Dapr, Apache Kafka, OCI OKE, PostgreSQL

AI Video Generation Agent — End-to-End Content Automation | [GitHub](#) | [YouTube Demo](#)

- Built complete AI pipeline: text prompt → script → storyboard → rendered 1080p video with synchronized narration
- Orchestrated LLM workflows using Claude Code (Opus) as orchestrator with Gemini TTS for voiceover generation

Tech: TypeScript, React, Remotion, Claude Code (Opus), Gemini TTS

AHF Auto Parts — Production E-Commerce Platform | [GitHub](#) | [Live](#)

- Built PostgreSQL-backed production platform with full admin analytics dashboard, order management, and inventory pipeline
- Implemented async data operations with 30-minute stock reservation (Upstash Redis) to prevent overselling under concurrent load

Tech: Next.js 16, TypeScript, PostgreSQL/Neon, Prisma, Stripe, Upstash Redis, Vercel

PROFESSIONAL EXPERIENCE

AI Engineer (Independent)

2023 – Present

- Building production AI systems, autonomous agents, data pipelines, and automation tools from research to deployment
- Conducting independent AI/ML research and translating innovative concepts into production-ready solutions
- Contributing to Pakistan's AI ecosystem through open-source projects and active PIAIC program participation

Quality Control Inspector & Electronics Technician (Retired) | Pakistan Air Force

February 2000 – March 2023 (23 Years)

- Led quality control inspection in Air Logistics and engineering depots under military-grade standards
- Managed cross-functional engineering teams; specialized in electronics and communications systems maintenance

EDUCATION & CERTIFICATIONS

- Certified Agentic and Robotic AI Engineer | PIAIC, Air University Islamabad | Expected 2026
- Master of Computer Science (CGPA: 3.11) | Virtual University of Pakistan | 2012
- DAE Radio Technology | School of Electronics, PAF Korangi Creek | 2003
- PCAP: Programming Essentials in Python | Cisco Networking Academy / OpenEDG Python Institute | 2022
- Artificial Intelligence and Computing Program | Presidential Initiative for AI and Computing | 2023

ADDITIONAL

- Veteran: 23+ years Pakistan Air Force — disciplined, independent, delivers under pressure
- Active open-source contributor: production AI agents and tools publicly available on GitHub
- Strong independent research skills — self-directed learner with proven ability to ship novel solutions
- Operating Systems: Linux Ubuntu, Windows